CDD with FTD. We also adapt the FTD [8] as the optimization objective for CDD as follows,

$$\begin{aligned} \mathop{\mathbb{E}}_{\theta\sim\theta_{1:N}}[\ell_{FTD}(\mathcal{S}_t, \{\mathcal{R}_t, \mathcal{H}_{1:t-1}\}; \theta)] &= \mathop{\mathbb{E}}_{\theta\sim\theta_{1:N}} \frac{\|(\theta_S^{(n)} - \theta^{(m)})\|_2^2}{\|(\theta - \theta^{(m)})\|_2^2}, \\ \theta_S^{(0)} = \theta,\ \theta_S^{(n)} &= \theta_S^{(n-1)} + \eta\nabla L(\mathcal{S}_t; \theta_S^{(n-1)}), \\ \theta^{(0)} = \theta,\ \theta^{(m)} &= \theta^{(m-1)} + \eta\nabla L(\{\mathcal{R}_t, \mathcal{H}_{1:t-1}\}; \theta^{(m-1)}), \end{aligned} \tag{9}$$

where N in Eq. 9 represents the total number of cached checkpoints θ in the history of learning task-t. The parameter vectors $\theta_S^{(n)}$ and $\theta^{(m)}$, which originate from the initial model parameters θ, are iteratively trained on the synthetic dataset $\mathcal{S}_t$ and the data $\mathcal{R}_t \cup \mathcal{H}_{1:t-1}$ of the current task, respectively, for n and m steps with learning rate η. The function $L(\cdot;\cdot)$ is the loss function used for training the network across different CIL methods. The objective in Eq. 9 focuses on quantifying the difference between two sets of parameter vectors: $\theta_S^{(n)}$ and $\theta^{(m)}$. Here $\theta_S^{(n)}$ is derived from the initial parameters θ, which are trained with synthetic exemplars $\mathcal{S}_t$ for n steps, while $\theta^{(m)}$ originates from the same initial parameters but is trained with real data $\mathcal{R}_t \cup \mathcal{H}_{1:t-1}$ for m steps.

CDD with DSA. Additionally, we also adapt the DSA [39] as the optimization objective for CDD as follows,

$$\mathop{\mathbb{E}}_{\theta\sim\theta_{1:N}}[\ell_{DSA}(\mathcal{S}_t, \mathcal{R}_t; \theta)] = \mathop{\mathbb{E}}_{\theta\sim\theta_{1:N}} \frac{< \nabla L(\mathcal{S}_t;\theta), \nabla L(\mathcal{R}_t;\theta) >}{\|\nabla L(\mathcal{S}_t;\theta)\|_2, \|\nabla L(\mathcal{R}_t;\theta)\|_2}, \tag{10}$$

where N in Eq. 10 represents the total number of cached checkpoints θ in the history of learning task-t. The function $L(\cdot;\cdot)$ is the loss function used for training the network across different CIL methods. The objective in Eq. 10 focuses on quantifying the cosine similarity between the gradient of the synthetic exemplars $\mathcal{S}_t$ and the gradient of the real dataset $\mathcal{R}_t$, both computed on the sampled parameters θ with the loss function $L(\cdot;\cdot)$.

CDD with DataDAM. Furthermore, we adapt the DataDAM [29] as the optimization objective for CDD as follows,

$$\mathop{\mathbb{E}}_{\theta\sim\theta_{1:N}}[\ell_{DataDAM}(\mathcal{S}_t, \mathcal{R}_t; \theta)] = \mathop{\mathbb{E}}_{\theta\sim\theta_{1:N}}[L_{SAM}(\mathcal{S}_t, \mathcal{R}_t; \theta) + \lambda L_{MMD}(\mathcal{S}_t, \mathcal{R}_t; \theta)], \tag{11}$$

where N in Eq. 10 represents the total number of cached checkpoints θ in the history of learning task-t. The objective in Eq. 11 focuses on attention matching and quantifying the distributions between synthetic exemplars and real datasets by objectives $L_{SAM}(\cdot,\cdot;\cdot)$ and $L_{MMD}(\cdot,\cdot;\cdot)$.

Furthermore, we validate the wild applicability of our CDD in adapting the data distillation methods into CIL. In this experiment, we adapt different DD algorithms, DM, FTD, DSA, and DataDAM above as CDD optimization objectives. These objectives with CDD are integrated into iCaRL, and their performance is evaluated on the CIFAR-100 dataset with zero-base 10 phases setting. In the experiment, each CDD is used to generate **10** synthetic exemplars per class, denoted as $S(DM)$, $S(FTD)$, $S(DSA)$, and $S(DataDAM)$. These are compared against the original iCaRL method, which selects 10 and 20 real exemplars per class using *herding*[34], as described in[25]. As illustrated in Fig. 7, iCaRL with 10 synthetic exemplars per class outperforms the original method which uses 10 real exemplars per class (black curve). In addition, using just 10 synthetic exemplars per class as $S(FTD)$ achieves comparable performance to iCaRL configured with 20 real exemplars per class (blue dash curve).

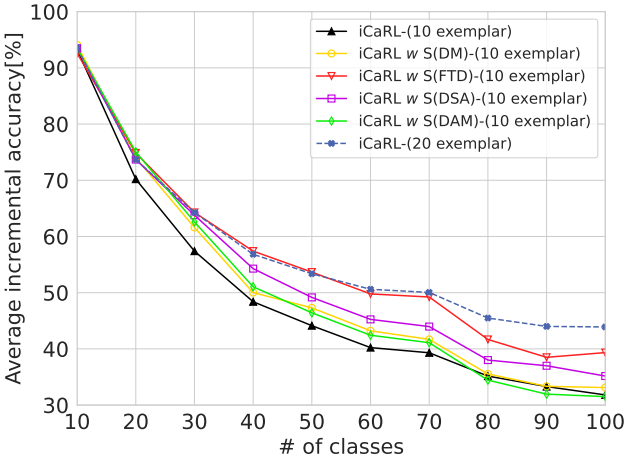
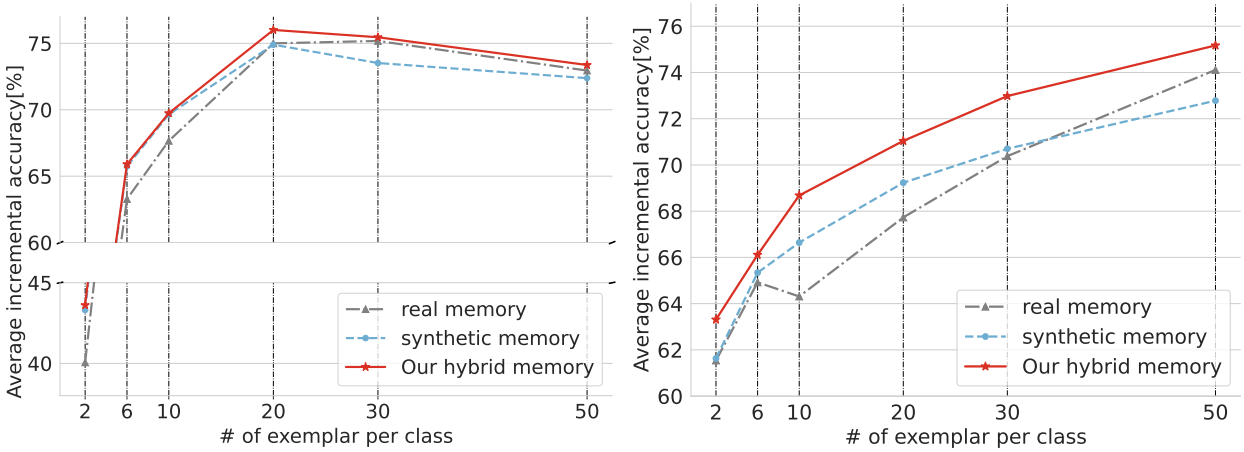


Figure 7: Results of different CDD optimization objectives on CIFAR-100 with zero-base 10 phases setting. S$(\cdot)$ means synthetic memory generated by a specific objective.

E Effectiveness of Hybrid Memory Across Different Exemplar Buffer Sizes

To further validate the assumption from Sec. 3 that synthetic exemplars alone are not always sufficient, we explore the effectiveness of the hybrid memory across different exemplar buffer sizes with several different replay-based CIL methods. In addition to the iCaRL results shown in Fig. 2, we compare the performance of hybrid memory with both synthetic and real memory on two other replay-based CIL methods, BEEF and FOSTER, across various buffer sizes.



(a) Performance of FOSTER (b) Performance of BEEF

Figure 8: Performance evaluations of FOSTER and BEEF across different exemplar buffer sizes for real memory, synthetic memory, and our hybrid memory. "real memory" refers to buffers containing only real exemplars selected by iCaRL. "synthetic memory" contains only synthetic exemplars generated using CDD.

The experiments were conducted on CIFAR-100 under the zero-base 10-phase setting, the same setting used for iCaRL in Fig. 2.

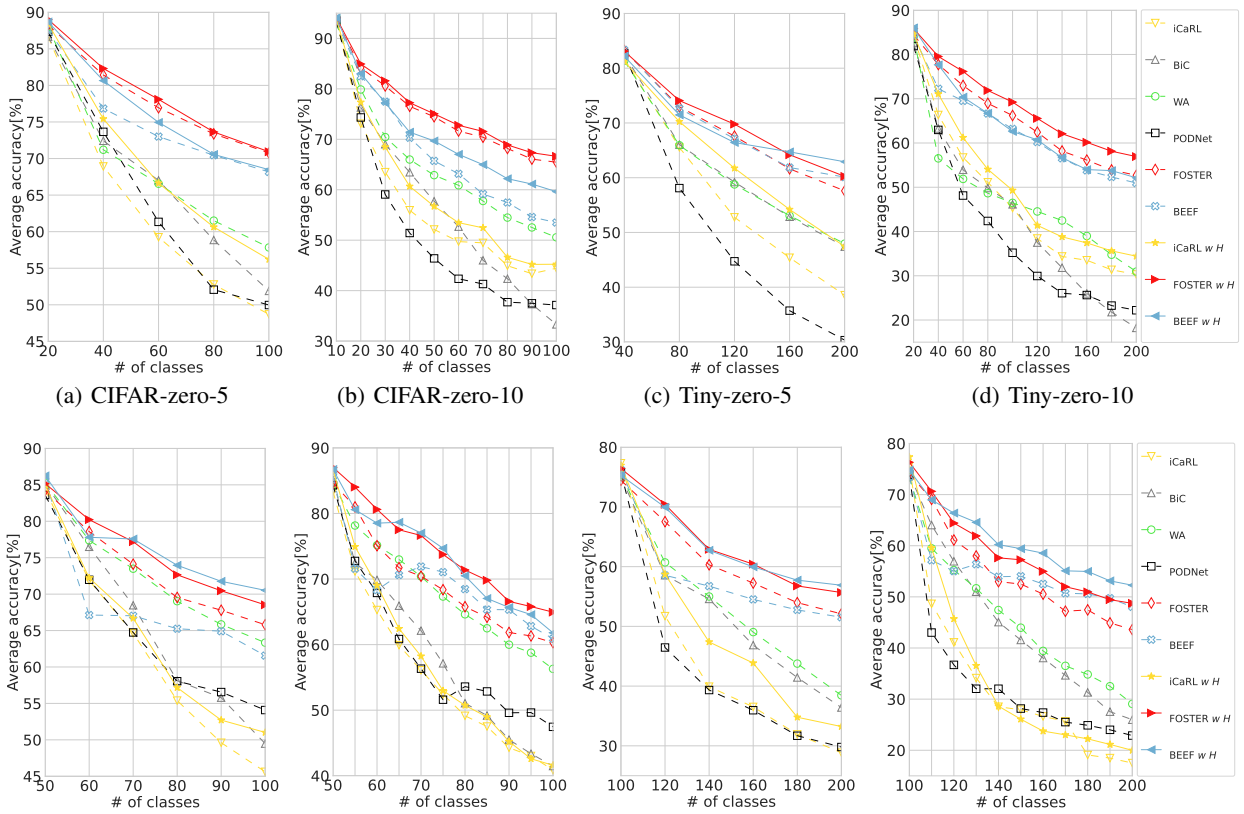
As shown in Fig. 8, the performance of synthetic memory generated by CDD significantly improves at smaller buffer sizes but quickly diminishes as the number of synthetic exemplars increases in both FOSTER and BEEF. In contrast, our proposed hybrid memory leverages the strengths of both synthetic and real exemplars, consistently outperforming synthetic and real memory across different exemplar buffer sizes.

F Additional Results

To demonstrate the superior performance of our approach, we also compare the average accuracy curves for each task against the baselines. As described in Sec. 5.1, we here present the average accuracy (AA) for each task corresponding to Table 1 and Table 2. As shown in Fig. 9, a comparison between the solid curves (representing methods with our proposed hybrid memory) and the dashed curves of the same color (representing the original methods) shows that our hybrid memory consistently enhances the performance of replay-based CIL methods, even with a limited exemplar buffer size, across different dataset scales and experimental settings.

G Visualization

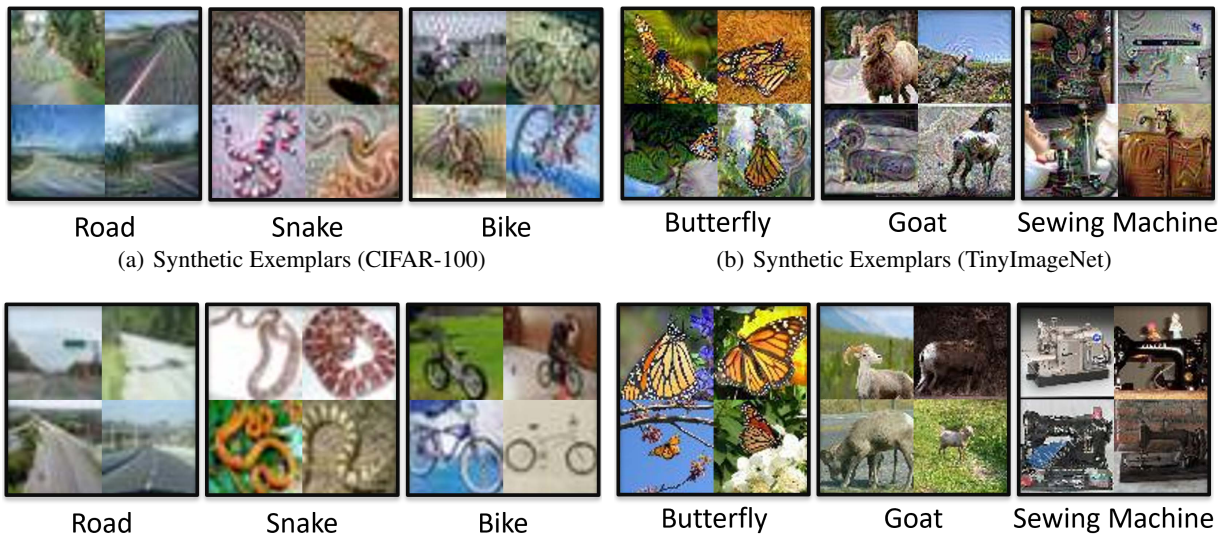
In this section, we provide visualizations of the synthetic and selected exemplars from CIFAR-100 and TinyImageNet. As shown in Fig. 10, we randomly select three different classes and compare the synthetic exemplars generated by our CDD with the corresponding selected exemplars conditioned on these synthetic exemplars. By comparing the upper and bottom rows, we observe that the selected exemplars are chosen based on the given synthetic exemplars and provide complementary information that the synthetic exemplars alone do not capture.



(a) CIFAR-zero-5 (b) CIFAR-zero-10 (c) Tiny-zero-5 (d) Tiny-zero-10

(e) CIFAR-half-5 (f) CIFAR-half-10 (g) Tiny-half-5 (h) Tiny-half-10

Figure 9: Accuracy comparison of different methods on CIFAR-100 and TinyImageNet under different settings. It can be observed that our hybrid memory can improve the performance of repla-based methods in terms of accuracy. “zero-5,10" means “zero-base-5,10 phases" setting and “half-5,10" means “half-base-5,10 phases" setting.



(a) Synthetic Exemplars (CIFAR-100) (b) Synthetic Exemplars (TinyImageNet)

(c) Select Exempalrs (CIFAR-100) (d) Select Exempalrs (TinyImageNet)

Figure 10: Visualization of select exemplars and synthetic exemplars, the latter generated by CDD with DM, from CIFAR-100 and TinyImageNet.